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## **CO1-AT3-ASSESSMENT TOOL**

### **1. Smart Home Automation System**

Types of Intelligent Agents

#### **1. Simple Reflex Agent**

Acts only on the current input using condition-action rules.

Example: If motion is detected → Turn ON the light.

#### **2. Model-Based Agent**

Maintains an internal model of the environment.

Uses past and current information.

Example: Knows whether a room is occupied even if motion stops.

#### **3. Goal-Based Agent**

Makes decisions to achieve a specific goal.

Example: Keep room temperature at 24°C while saving energy.

#### **4. Utility-Based Agent**

Chooses the action with the highest overall benefit.

Balances comfort, energy efficiency, and security.

Example: Adjusts AC, lights, and security to maximize comfort while reducing electricity usage.

How They Work in the Smart Home

Agent Type

Function

Simple Reflex

Turns lights ON/OFF based on motion.

Model-Based

Remembers room occupancy and user habits.

Goal-Based

Maintains desired temperature and security.

Utility-Based

Selects the best action by balancing comfort, energy, and safety.

Best Choice

A Hybrid Agent (Model-Based + Utility-Based) is the most effective because it:

Learns user preferences.

Responds to real-time sensor data.

Saves energy.

Improves comfort.

Provides better security.

Conclusion:

Hybrid (Model-Based + Utility-Based) is the best choice for a smart home.

## 2. Robot in an Unknown Maze (10 Marks)

Uniform Cost Search (UCS)

Expands the path with the lowest cost first.

Finds the optimal (shortest-cost) path.

Advantages

Complete.

Always finds the least-cost path.

Disadvantages

Uses more memory.

Slower for large mazes.

Iterative Deepening Search (IDS)

Repeats Depth-First Search with increasing depth limits.

Finds the goal without large memory usage.

Advantages

Low memory requirement.

Complete for finite search spaces.

Disadvantages

Repeats searches many times.

Slower because of repeated exploration.

Time and Space Complexity

Algorithm

Time

Space

UCS

Exponential (depends on path costs)

High

IDS

$O(b^d)$

$O(b \times d)$  (Low)

( $b$  = branching factor,  $d$  = depth of solution)

Relation to Intelligent Agents

Simple Reflex Agent: Cannot solve the maze because it only reacts to the current state.

Model-Based Agent: Builds a map of explored paths and avoids revisiting locations.

Goal-Based Agent: Searches for the exit using a search algorithm.

Utility-Based Agent: Chooses paths that minimize time, energy, or risk.

Best Combination

Goal-Based Agent + Uniform Cost Search (UCS)

Reason:

Goal-Based Agent focuses on reaching the exit.

UCS guarantees the lowest-cost (best) path.

If memory is limited, Goal-Based Agent + IDS is a good alternative.

### Conclusion:

For an unknown maze, Goal-Based Agent with UCS is the most effective because it reliably finds the optimal path, while IDS is preferred when memory is limited.